DIPLOMA WING
RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA, BHOPAL
DIPLOMA IN COMPUTER SCIENCE & ENGINEERING (C04)

SEMESTER VI

COURSE TITLE	:	FREE AND OPEN SOURCE SOFTWARE(FOSS)
PAPER CODE	:	7500
SUBJECT CODE	:	622
TREORY CREDITS	:	03
PRACTICAL CREDITS	:	02

Course Learning Objectives:

Exposure to free and open source software philosophy and tools.

Course Content:

UNIT 1: FOSS PHILOSOPHY

Understanding the FOSS Community and FOSS Philosophy, Benefits of Community based Software Development, Guidelines for working with FOSS community, Requirements for being open, free software, open source software, FOSS Licensing Models, FOSS examples

UNIT 2: LINUX

Linux Installation and Hardware Configuration, Boot Process, Dual-Booting Linux and other Operating Systems, Kernel Options during Boot, X Windows System Configuration, System Administration (Server Administration, Backup and Restore Procedures, Strategies for keeping a Secure Server)

UNIT 3: Programming Tools and Techniques

Libreoffice Tools; Samba: Cross platform; Introduction about LAMP; Brief Introduction to Programming using languages like Java /Python / Perl; Database Systems Mysql, PostgreSQL or equivalent; Open Source UML Tools; Introduction to Mobile Programming; Version Control Systems like SVN, Git or equivalent; Project Management Tools; Bug Tracking Systems; Package Management Systems

UNIT 4: FOSS CASE STUDIES

Some example case studies of FOSS implementation

Suggested Lab Work:

This is a skill course. Topics/tools taught in the class should be practiced in the Lab same week and practiced regularly during the semester till student becomes confident about it. Students should explore features of various FOSS tools/applications on a Linux system. Teacher should give weekly tasks as assignment. Learnings from this course should be used in the major project.

Reference Books:

8. **Linux in a Nutshell, by Ellen Siever**
9. Philosophy of GNU URL: http://www.gnu.org/philosophy/.
10. Linux Administration URL: http://www.tldp.org/LDP/lame/LAME/linux-admin-made-easy/.
11. Version control system URL: http://git-scm.com/.
12. Samba: URL : http://www.samba.org/.
13. Libre office: http://www.libreoffice.org/.

Course outcomes:

Student will be able to work with FOSS tools, find and evaluate FOSS alternatives for any software requirement.